RNA-seq Differential Expression Analysis of Type 2 Diabetes Mellitus vs Healthy Controls

Independent computational reanalysis | Research portfolio report

PROJECT SCOPE • Public human bulk RNA-seq data • 5 T2DM samples and 5 controls • Galaxy-based analysis and focused functional interpretation

ABSTRACT

We conducted an independent reanalysis of ten publicly available human RNA-seq runs (five Type 2 Diabetes Mellitus [T2DM] and five controls) selected from a larger 22-sample dataset associated with Gene Expression Omnibus (GEO) accession GSE326582, identified by the study title “Transcriptome Profile of Pancreatic Adipose Tissue in Pancreas.” Paired-end reads underwent quality assessment, preprocessing, splice-aware alignment to hg38, gene-level quantification, and differential expression analysis with DESeq2. Fifty-seven genes met an adjusted p-value threshold of 0.05. Gene Ontology enrichment highlighted extracellular matrix (ECM)-associated processes, motivating focused assessment of SFRP2, CCDC80, CTSK, OLFML2B, FBLN1, LOXL1, and COL1A1. All seven showed higher RNA expression in T2DM than controls (T2DM-versus-Control log2 fold changes approximately +1.40 to +1.97; adjusted p-values < 0.05). These exploratory results suggest ECM-associated transcriptional differences in the selected samples, without establishing causality or protein-level changes.

RESEARCH OBJECTIVE

Identify differentially expressed genes between selected T2DM and control RNA-seq samples, assess enriched functional categories, and characterize expression and known functions of seven ECM-associated genes.

DATA SOURCE AND SAMPLE SELECTION

The source dataset is identified by GEO accession GSE326582 and the supplied study title “Transcriptome Profile of Pancreatic Adipose Tissue in Pancreas.” The parent dataset contains 22 samples; this independent analysis used a subset of ten runs selected for a five-versus-five T2DM/control comparison. The full dataset was not analyzed, and the remaining samples should not be assumed to share the same condition or characteristics. The original publication citation, detailed tissue descriptions, and individual clinical covariates were not independently verified for this report; they should be confirmed from the GEO record before external scientific submission.

T2DM (n = 5)	Controls (n = 5)
SRR37874037	SRR37874054
SRR37874038	SRR37874055
SRR37874039	SRR37874056
SRR37874040	SRR37874057
SRR37874041	SRR37874058

DATA AND METHODS

Read processing. Raw paired-end FASTQ files were assessed with FastQC, trimmed/filtered using fastp, and reviewed again following preprocessing; MultiQC supported consolidated quality review. Run identifiers and read pairs were tracked to maintain sample identity.

Alignment and quantification. Cleaned reads were mapped to the hg38 human reference using HISAT2, and gene-level counts were generated with featureCounts. Count columns were matched to sample conditions for downstream analysis.

Differential expression. DESeq2 was used to compare the five selected T2DM and five control samples, with adjusted p-value < 0.05 defining statistical significance. The original output contrast was Control versus T2DM. For consistent reporting here, log2 fold changes were sign-reversed to represent T2DM versus Control; adjusted p-values were unchanged.

Functional analysis and visualization. Significant genes were assessed using g:Profiler Gene Ontology enrichment. Principal component analysis (PCA), MA/volcano plots, and heatmaps supported exploratory interpretation. Seven ECM-associated genes were annotated using gene symbols and general biological functions. A row-scaled variance-stabilized-transformation (VST) heatmap compared their expression across the ten samples.

RESULTS

Differential expression. Fifty-seven genes were significant at adjusted p-value < 0.05; in the T2DM-versus-Control direction, 47 showed higher expression in T2DM and 10 showed higher expression in controls.

Exploratory patterns. PC1 and PC2 accounted for approximately 31% and 20% of variance, respectively, without clear complete separation between groups.

Functional enrichment. Overrepresented categories included extracellular matrix organization, cell adhesion, extracellular region, fibronectin binding, and integrin binding. Enrichment describes association among the significant genes, not confirmed pathway activation.

Focused ECM-associated genes. All seven selected genes had positive T2DM-versus-Control log2 fold changes and adjusted p-values below 0.05. The row-scaled VST heatmap showed relatively higher expression of these genes in the T2DM samples.

Gene	log2FC (T2DM vs Control)	Adjusted p-value
SFRP2	+1.974	1.83×10^{-6}
CCDC80	+1.855	4.13×10^{-5}
CTSK	+1.477	0.01469
OLFML2B	+1.403	0.01738
FBLN1	+1.408	0.02005
LOXL1	+1.636	0.04564
COL1A1	+1.682	0.04573

DATA VISUALIZATION

Figures below summarize exploratory sample variation, differential expression, and the focused ECM-associated gene analysis.

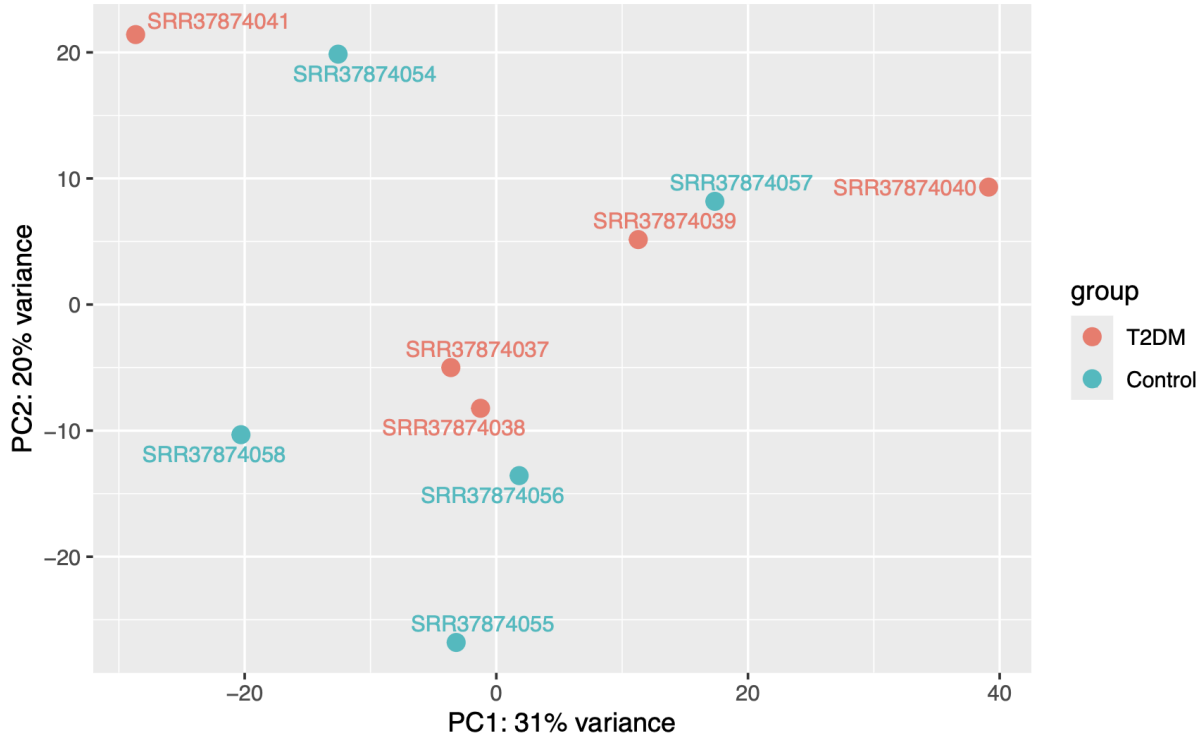


Figure 1. Principal component analysis (PCA). PC1 and PC2 explain approximately 31% and 20% of variance; the groups do not show complete separation.

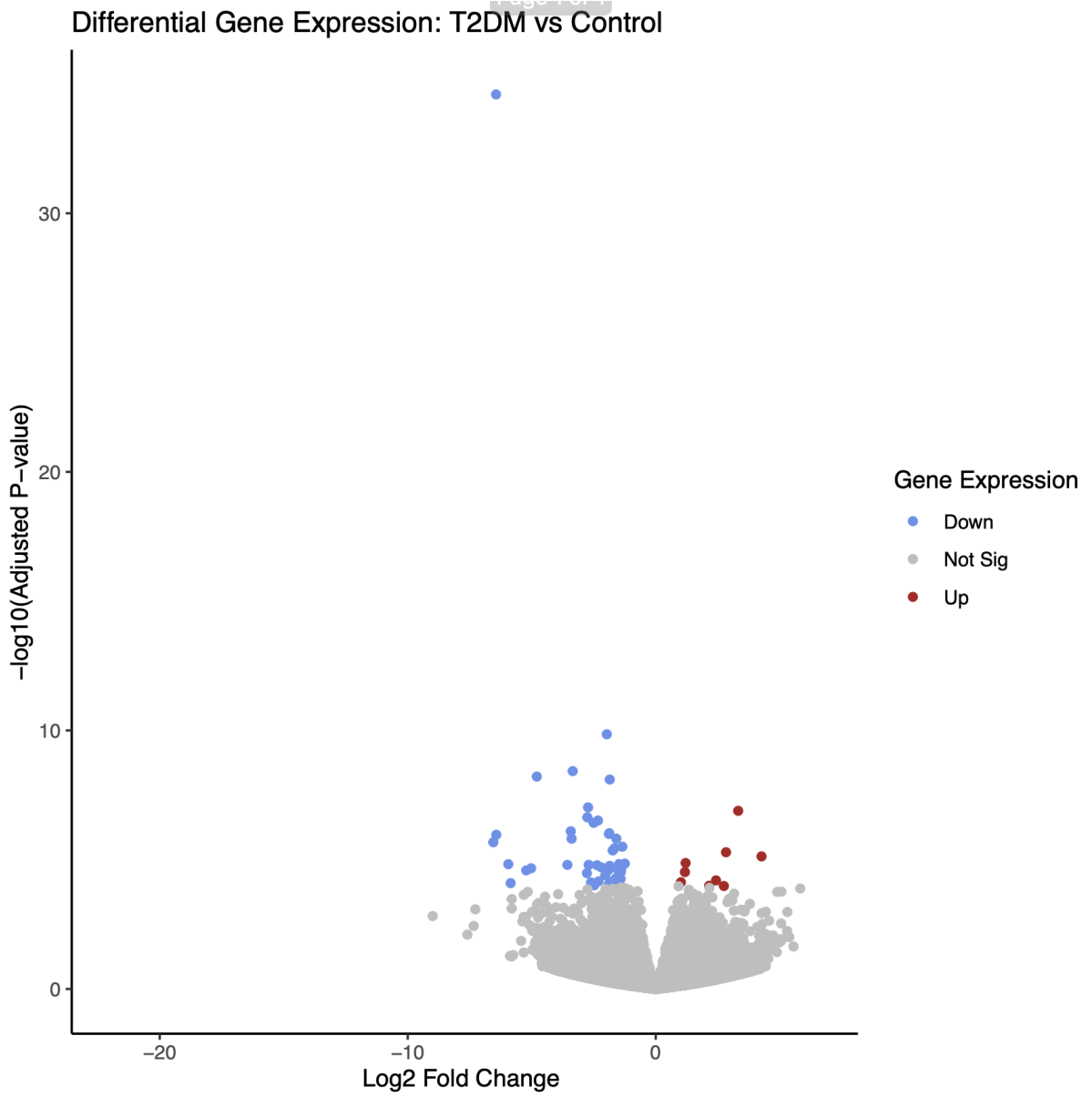


Figure 2. Volcano plot of differential expression results. Significance and fold-change patterns are shown for the ten-sample comparison.

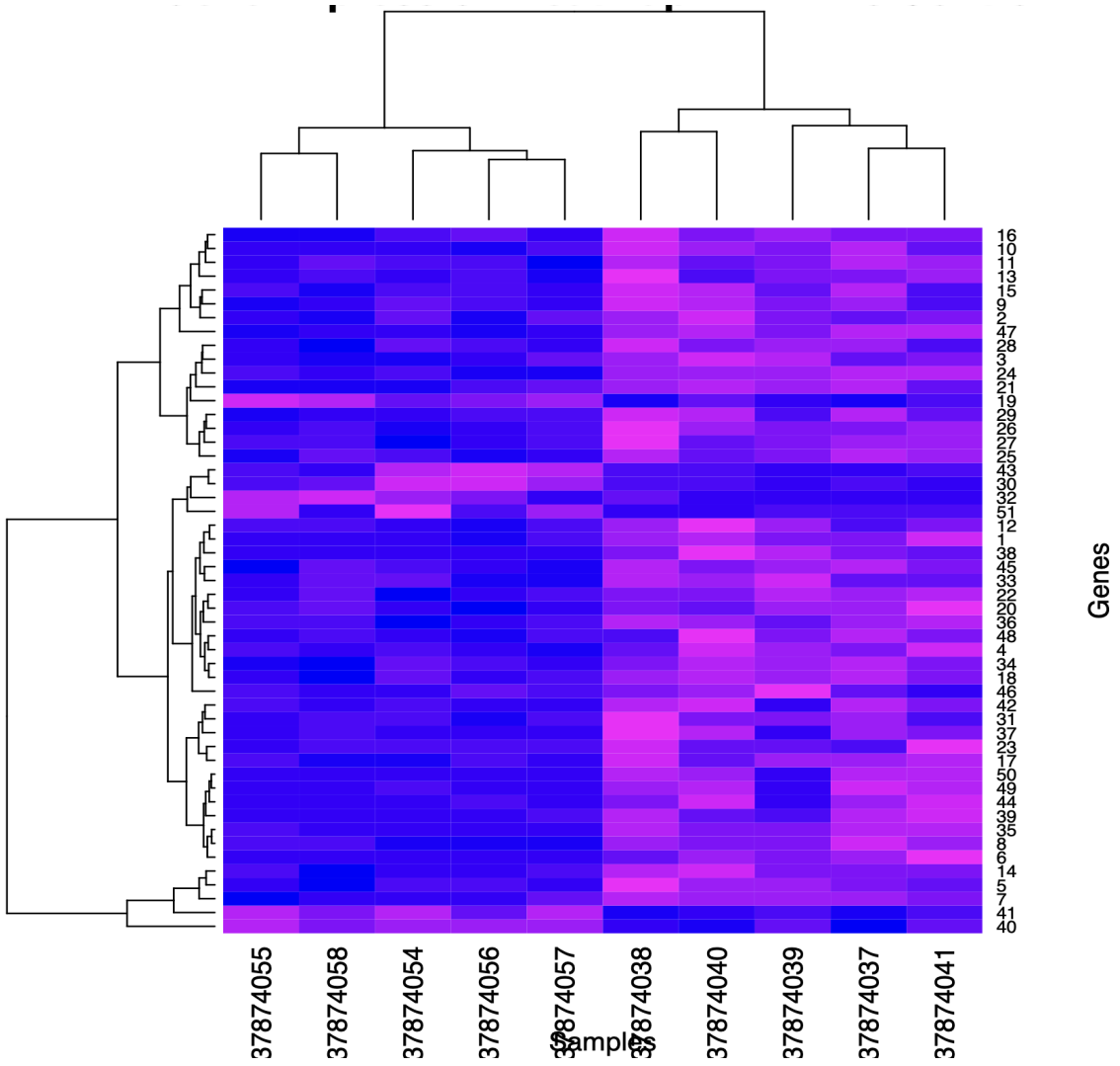


Figure 3. Expression heatmap from the initial RNA-seq analysis. Colors indicate relative expression patterns, not direct measures of protein abundance.

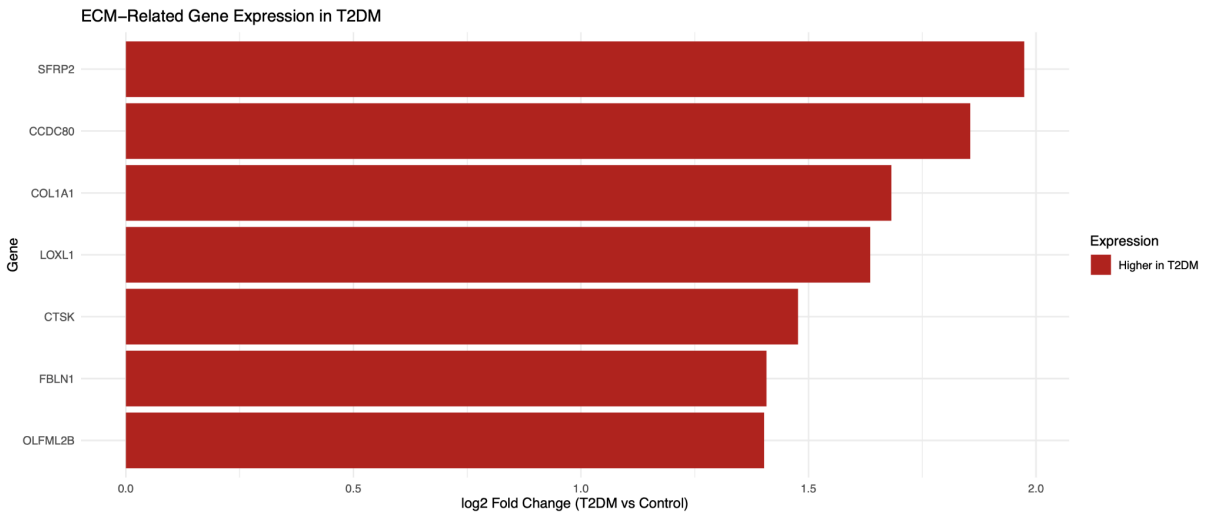


Figure 4. Log2 fold changes for seven ECM-associated genes. Positive values indicate higher expression in T2DM versus control.

BIOLOGICAL INTERPRETATION

The selected genes have functions relevant to extracellular biology: COL1A1 encodes a chain of type I collagen; FBLN1 is an ECM glycoprotein; LOXL1 contributes to elastin cross-linking; CTSK encodes a collagen-degrading protease; SFRP2 modulates Wnt signaling; and CCDC80 and OLFML2B encode extracellular or ECM-associated proteins with less fully characterized functions. Their shared direction of differential expression motivates further investigation of ECM-related tissue responses in this dataset. RNA expression alone cannot demonstrate ECM deposition, fibrosis, enzyme activity, or changes in Wnt pathway activity.

LIMITATIONS

The analysis is limited to five selected runs per group from a 22-sample parent dataset, and the selection may not capture the full study design or biological heterogeneity. Tissue characteristics, covariates, technical batch effects, and the source publication were not independently confirmed. PCA did not show strong group separation. No independent cohort, wet-lab validation, protein-level assay, or causal experiment was performed. Gene Ontology enrichment is associative, and the focused seven-gene analysis is exploratory.

CONCLUSION

In this ten-sample reanalysis of GSE326582, 57 genes were differentially expressed between T2DM and controls, and functional enrichment highlighted ECM-associated biology. Seven selected ECM-associated genes showed higher RNA expression in T2DM. These findings provide a hypothesis-generating basis for independent replication and mechanistic investigation.

PROJECT OUTPUTS AND REPRODUCIBILITY

Outputs include QC summaries, alignment and count results, DESeq2 tables, PCA/MA/volcano visualizations, functional enrichment results, an ECM fold-change plot, a seven-gene VST heatmap, and annotated results tables. Full reproducibility would benefit from sharing the analysis scripts, exact tool versions and parameters, reference annotation release, and validated sample metadata alongside the report.

DATA AND TOOL REFERENCES

Dataset: NCBI Gene Expression Omnibus (GEO), accession GSE326582; selected Sequence Read Archive (SRA) runs listed above. Study title supplied by project author: “Transcriptome Profile of Pancreatic Adipose Tissue in Pancreas.” GEO record: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE326582. Analysis tools: FastQC, fastp, MultiQC, HISAT2, featureCounts, DESeq2, g:Profiler, NCBI Gene, and R. The original study’s bibliographic citation should be added after verification.